MERN STACK

DAY – 33

1 August 2026

PROJECT: PARKEASY - PARKING MANAGEMENT SYSTEM

USER DASHBOARD DEVELOPMENT

• The User Dashboard is developed to provide users with their personal parking information and activities.

• It allows users to view vehicle details, booking history, parking status, and available parking slots.

• The useState() hook is used to store user dashboard data received from the backend.

• The useEffect() hook is used to fetch user information when the dashboard loads.

• The user ID is retrieved from localStorage after successful login authentication.

• The User Dashboard fetches user details from the backend using the API:

```
GET /user-dashboard/:id
```

• The dashboard displays:

1. User Name
2. Email Address
3. Vehicle Number
4. Vehicle Type
5. Current Parking Status
6. Total Bookings
7. Available Parking Slots

• The active booking is identified by checking:

```
booking.status === "Approved"
```

• Quick action buttons are provided for:

- Book Slot
- Available Slots
- My Bookings

• The dashboard also displays account information including:

- Name
- Email
- Vehicle Number
- Parking Status

USER DASHBOARD API DEVELOPMENT

• The User Dashboard API is created using Express.js to retrieve user-specific information.

• The API uses the logged-in user's ID to fetch data from MongoDB.

API Endpoint:

```
GET /user-dashboard/:id
```

• User information is retrieved using:

```
User.findById(req.params.id)
```

• Available parking slots are counted using:

```
Slot.countDocuments({

status:"Available"

})
```

• The API returns:

- User Details

- Vehicle Information
- Booking Records
- Available Parking Slots

• The response is sent to the frontend in JSON format.

Working

1. The UserDashboard() component is created to display user information.
2. The user ID stored in localStorage is retrieved.
3. A request is sent to the User Dashboard API.
4. The backend searches the user details using the provided ID.
5. Available parking slots are counted from the Slot collection.
6. The API returns user details and booking information.
7. The frontend displays parking activities and account information.
8. Users can navigate to booking-related pages using quick action buttons.

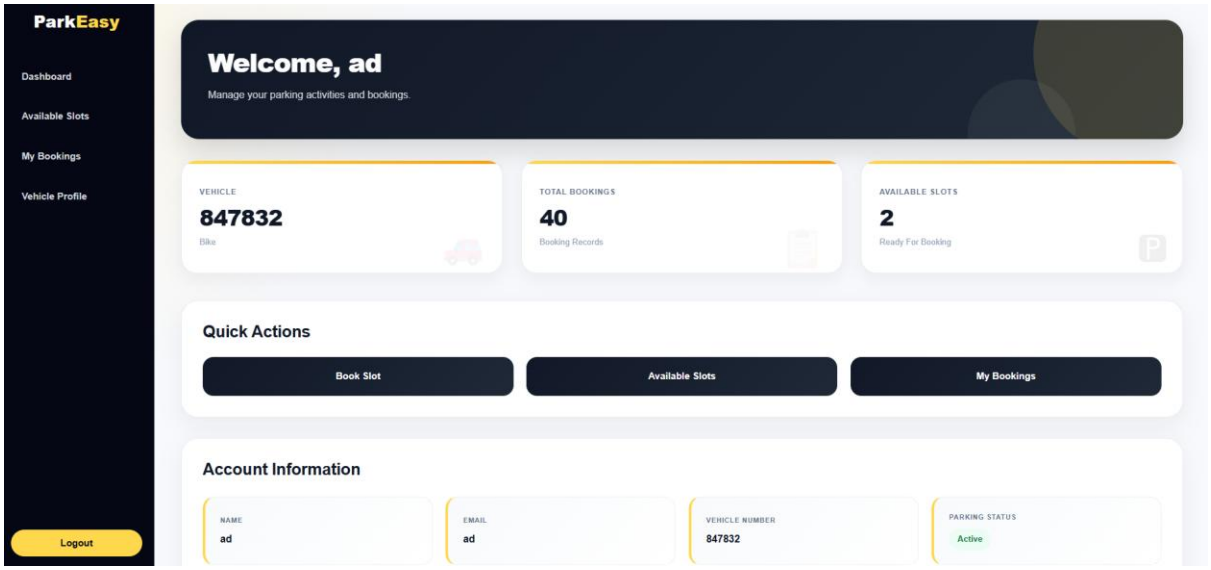


```
app.get("/user-dashboard/:id",async(req,res)=>{
    try{
        const user=await User.findById(req.params.id);

        const availableSlots=await Slot.countDocuments({
            status:"Available"
        });

        res.json({
            success:true,
            name:user.name,
            email:user.email,
            vehicle:user.vehicle,
            bookings:user.bookings,
            availableSlots
        });

    }catch(err){
        res.json({
            success:false,
            message:err.message
        });
    }
});
```


ParkEasy
Dashboard
Available Slots
My Bookings
Vehicle Profile
Logout
Welcome, ad
Manage your parking activities and bookings.
VEHICLE
847832
Bike
TOTAL BOOKINGS
40
Booking Records
AVAILABLE SLOTS
2
Ready For Booking
Quick Actions
Book Slot
Available Slots
My Bookings
Account Information
NAME
ad
EMAIL
ad
VEHICLE NUMBER
847832
PARKING STATUS
Active